

ActInf Livestream #013.1: "Cybernetic Big Five Theory with the Free Energy Principle..."

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hello everyone and welcome to the active inference Liv stream this is active inference Liv stream 13.1 and it is January 5th 2021 happy New Year everyone welcome to the active inference lab the active inference lab is an experiment in online team communication learning and practice related to active inference feed back you can find us at our website Twitter email YouTube or keybase teams this is a recorded and an archived live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome here and keep in mind video etiquette for live streams for 2021 we are going to be meeting every Tuesday regularly at 7: to 9:00 a.m. pacific time and then we'll plan other special sessions as things come up and for different time zones you can go to this link and see a spreadsheet and here we are in the first week with the first paper and next week we'll be discussing the same one so hello everyone thanks for coming on live and thanks everybody who's watching let's introduce ourselves please give a short introduction or check-in especially if it's your first time on the stream and then pass it to somebody who hasn't spoken so I'm Daniel I'm in Davis and I'll pass it to Colin I'm Colin D young and I'm in Minneapolis Minnesota um I'm a professor in the psychology department here at the University of Minnesota and excited to be here um I'll pass it on to Stephen oh hello I'm Stephen I'm based in Toronto um I'm doing my practice based PhD at Canary Christ Church University and uh pleased to be here as well pass it to Shannon hey I'm Shannon PRS I'm in South Dakota so also in the snow but I'm based at the University of California in Merced during my PhD also I'll pass it to Marco hi I'm Marco um I'm from Holland and used to work on active inference in London but now I'm just um enjoying the progress from the sidelines I'll pass it to uh Ean thank you my name is Ian I'm from Russia from Moscow and I'm interested in free energy principle and as it related with systems engineering and I pass it to uh excuse me what is your name my friend Dave let's go to Dave hi Dave Douglas I'm retired from Information Technology I live in the Philippines and I did about equal amounts of graduate level work in computer science Linguistics and psychology

but have only a lowly ba cool let's go to blue and then Adam hi good morning everyone my name is blue Knight and I am an independent research consultant based out of New Mexico and Adam welcome hi I'm Adam sorry I'm late I am based out of Bloomington Indiana uh I'm a postural researcher cool well welcome everyone thanks so much for coming on filling out the Hollywood Square that we have here for these warm-up questions it'll be great to hear from everyone just raise your hand if you have a thought I'll just put up the first two totally just raise your hand as you want to add something first would be what is something about human personality or emotions that you would want to see captured in a good model or understanding or education and then also what is something that you liked or remembered about the paper and um I think there was a few memorable parts of the paper just while people are raising their hand uh I think just the idea of going from a statistical understanding of the big five as principal component Dimensions to being the outputs of a cybernetic system is really an interesting topic so maybe if uh anyone wants to raise their hand on this one or Adam or Colin maybe how do you go about introducing uh the the need for your kind of paper to an audience like where do you start out with the interesting part about personality that you want to capture in your model well uh shall I jump in then um and when you say raise hand do you mean this little button here or you like actually just wave at the screen probably with the button all right well works that works um so uh one of the things that I noticed in your summary of the paper uh was something that I often encounter from people working in other scientific Fields uh which is a certain degree of skepticism about the big five because of their origin in uh natural language and um and because you know when you see a diagram that's like a hierarchy of personality traits I you know in the figure there I was careful to note that that is not supposed to be a diagram of causal structure in the individual because there's this interesting problem where we use the same visual language for causal models of you know boxes and circles and arrows uh and for statistical models in which we're looking at the relationships among variables but obviously those are not directly translatable and um so the way that I would pose the basic question that's of Interest here is uh why do certain traits of people tend to Cluster together I think that that's really the core of what I'm interested in and what we're interested in if we're talking about the sources of Personality a lot of Personality research in the last 40 years or so since the big F emerged as kind of an organizing consensus framework has really been focused on questions of prediction like if we have these assessments of people's personality Tendencies what kind of outcomes does it predict for them um also potentially like what are trajectories of uh change and development of these things

over time but I'm interested in a question that is a lot less studied which is where do these patterns come from in the first place um and so I think the way that to think about the kind of statistical element here is simply that there is an observation that if we have a large set of descriptions of uh people's personality and it doesn't have to be from the natural language that was just one good source for a relatively unbiased set of descriptions of people was to go into the dictionary and just pull out the adjectives that could describe people um but we see very similar pattern of factors like the big five if we start with a lot of say existing personality questionnaires that weren't designed to measure the big five you throw them all together you still get the big five um so as long as we have a sufficiently large set of descriptions of personality and we have ratings on them uh they tend to cluster together in reliable ways and um as long as they are valid you know as long as when we say that somebody's talkative that really does mean that they talk more than other people do and there's a lot of evidence for the basic levels of validity of Personality ratings then the question is why is it that people who are talkative also tend to be uh more sociable also tend to experience more positive emotions also tend to be more physically active also tend to be uh let's say more interested in excitement and uh dangerous you know risky Pursuits so we've got a whole bunch of things that cluster together and the question is why and so you know that was a list of low-level traits that are grouped together within extra version and that's just a statistical observation those people who tend to have more of one of those things also tend to have more of another and so I see fundamentally the question that I'm interested in is what are the underlying mechanisms uh that vary from person to person um that drive these patterns of co-occurrence of particular characteristics thank you for that Shannon thanks for that um really nice sort of overview introduction um I was sort of wondering going through this paper you talk about what are the differences that are um can be ascribed to individuals and their individual personalities that lead to you know traits like extraversion or what have you um but what if there is nothing unique to any individual that gives rise to a personality trait and I wonder where does um culture and social interaction fit into this kind of model and that's just a question that I'm interested in throughout the next couple of hours that we're here today nice good question anyone I put up also the third one blue so something as I was wondering about like as I was reading the paper was you know the Big Five personality and like it's um you know grounded in genetic uh influences right um but like attachment Theory and like there's like a bunch of work that's been done in mice like showing like you know the nurturing of parents and you know whether you're attachment avoidant or attachment anxious or um you know attachment

secure that's also something I think would be fun and interesting to um model in in the free energy kind of way like put into into this framework and I think it has like some cool um you know impacts for like team building and and how um you know like whether I don't know I just think it would be it would be interesting and maybe um impactful in in this kind of framework cool and blew to that point I just saw a paper this uh morning from Anna Cuna at all an axle constant the first prior from co- embodiment to co- homeostasis in early life about this transition from like a truly in utero embedded being to different types of physiological dependencies social and cultural so it is interesting to think about all right we have an individual orientation around the big five it's about the individual in relationship to different types of settings but then how do you capture that element of Niche construction or of group dynamics or of these other aspects that are relational I'm I'm just curious Colin or or Adam like where does the uh other how how does that enter into it this idea of the cultural influences in on personality because someone's personality might be manifested in a really specific setting because of their local context I could just say something about that that is relevant I think to both of the two questions that have just been asked uh which is just to keep in mind that U these personality traits are descriptions of uh you know persistent patterns of people people's behavior and we know that they are genetically influenced but we also know that they're not entirely determined by genetics they are also determined by environmental influences and so the way that I think about this is that you have these uh you know these mechanisms that are allowing people to function and some parameters of those mechanisms must be reasonably persistent to produce these persistent patterns of behavior but the the things that set those parameters and that tweak them if there's change in personality over time can be both genetically programmed and environmentally programmed right so there are things there are events that happen to people for example their relations to their primary caregiver and attachment um that shape what these settings are going to be persistently in life um and so I think there's definitely room for thinking about environmental inputs uh to these systems and shaping of these parameters uh as well as genetic ones cool and Colin this point you raised earlier about uh graphical models and just drawing a connection among different variables is that a causal model is that a statistical model are those different what if you're talking about causal entropy or you're doing a causal uh you know model like a structural equation modeling or even if it's a linear aggression it's going to be interpreted as a causal influence and they'll put in the discussion oh it could be a different way it could just be correlation but it's really hard to disentangle that um yeah Marco hi thanks um so first of all um I haven't fully read the

paper I kind of recently got out of a pandemic um Slumber um but but it sounds very interesting from what I've read and heard so far and as far the first question about what I'd like to see it might be a bit um in the far future but I think ideally um this would all connect sooner or later to the work on neurop phenomenology the kind of Ideal Vision I have is for example now we talking about um genetic prior and environmental inputs but I think the beautiful part about cybernetics is the kind of reflex of action and so Consciousness or phenomenology is basically a mediator or moderator of the way that environmental inputs kind of influence or continuing progression of Personality or um traits and and characteristics and so ideally um I would love to see a well I would love to hear if it's possible to make a synthesis of for example the work from psychoanalysis Psychotherapy everything that kind of uh explores and navigates these particular mechanisms from a first- person kind of view how to navigate the experience of being engaged in the world and I'm just wondering about Adams or col someone else's thoughts on how to possibly connect that phenomological um description of mechanisms and processes with the more formal um basic models that we're now exploring in this kind of work if that makes sense yeah very nice Adam and then anyone else uh hello can you hear me yep great um so I guess to uh try to address a couple things um so yeah uh Marco just thrust us into like the deepest possible territory like right away um and so I guess I'll briefly comment on that uh Second but the I guess the first thing in terms of um social context and uh Niche construction and the more inter subjective aspects and context um sensitivity of Personality uh active inferential modeling is really well equipped to uh address all of that uh in this paper it was more uh the focus was just kind of exploring what K and I both perceive to be a very rich set of correspondences between between the free energy principle and active inference and uh his theory of cybernetic Big Five Theory but in um modeling work you can uh bring in all of those details and explore them in their particulars so I'm currently collaborating on a project where we're trying to model things like extroversion within these um social Choice tasks and so how you show up it's going to be a function of you know both the task environment and you and this interaction that's going to evoke a given style pattern but you'll have these um parameters uh these hyper priors are um over the different um aspects of the generative model which through iterated policy selection and uh updating will basically concentrate their probability mass in different ways and result in different parameterizations and then different emergent behavior and what you can see happen is you can get uh these basically attractors form as this process proceeds over time these behavioral attractors where you the concentration of your prior for engaging in the world in a certain way causes you to enact a

given pattern which causes you to uh even more update in that direction like your your D Matrix like where am I in this environment uh you know what do you expect to be perceiving your a matrices all the different like components that You' use to try to model the parts of an agent these were all your prior over these all get updated will then cause the pattern of in action so on and so forth and that over time this could cause you to be in this attractor where you have this enduring stability um how enduring you know it would depend on a variety of things uh and but so I guess that's one so in this process uh the social context who you're interacting with that would make make all the difference in terms of like how the kind of ball rolls down this epigenetic landscape like how you end up like carving out a given pattern for this agent uh with respect to uh neurophenomenology and just the uh actually getting it the richness of experience there's a sense in which all active influential modeling is already in a way um kind of personality modeling it's like you're kind of getting into like um uh these enduring um you these cybernet what they're thinking them as these like cybernetic um parameters for this fundamental process um or just the your ability to describe um the essence of the system uh the it's its normal form description this this maximally encompassing uh description of the modes of where you expect Define the system as a kind of probabilistic density in some kind of multi-dimensional phase space um if you have such a a attracting Network structure that you can describe in a way you could call that the personality of the system um but cybern big five Theory um kind of goes further and and or it it's grounding this in terms of well why should we expect certain modes over others why should we expect this then relates it back to the fundamental cybernetic cycle of of goal Pursuit that any agent has to do um so with respect to neurop phenomenology and it seems like the attempts to get at this within active inference things like uh sophisticated effect of inference the the models of uh metacognition where they're beginning to Model A kinds of mindful processing of reflecting on your um your confidence and and then this influencing your uh you you have having an actual model of your epistemic status you reflect you reflecting on this and this influencing how you sample influencing uh how confident you are in different ways um influencing What policies you end up deploying uh these things to the extent that they allow for um phenomenological richness they could be brought into an agent model and you're beginning to do a connection between neurop phenomenology or a process theory of that would dock of phenomenology and personality modeling so those are some thoughts nice thanks and to add on that Adam because that was really well said it's almost like some of these past papers we've been discussing were the neutral infrastructure or the architecture or the tools to have agents

that can have memory or can have affect or can perceive veillance or get worried so we talked that in a very neutral framework like it's a parameter in the model however in the real world as Colin was getting at there's these extremely deep correlations with different outcomes of systems people who like to text more also like to video chat more or something and that might not be surprising but it these Trends are very distinct and so we are getting at this generative model idea personality as a generator of outcomes in different niches so Inseparable from the niche and um it's just an interesting way to see how the cybernetic big five has come from a more psychological area so it has uh that sort of um toe of the neurop phenomenology and the real correlations in the world and then there's the neutral framework that sort of maps onto that we have Steph and then Colin yeah I like what's been said about neurophenomenology and I'd also say that one of the things that's good about the this this work with traits is it it's starting to bring work that's coming from um what you could say realist settings codifying it and then finding the patterns and it's given us more U material coming the other way we've had some work around a lot of work with affordances and niches and then obviously you've got the models which start to create Dynamics and show and then you start to see where these two things meet but I would also say micro I've been looking at lot of micr phenomenology and the actual structuring of experience and attention and I think that um what can be interesting is these traits if these traits can also get brought down into micr phenomenology and ways of like how is the structure of the environment body mind um happening and and where are their times because what I think with a lot of the trait works is it does have a normative assumption about the type of Life someone's in around which you then standardize and then get these patterns but you do find if you take people into some other types of space or other types of attentional State be it a more grounded State maybe a a performative state maybe a a space Which is less Western so to speak less goal orientated um that might shift so this might be like this is where that is when people are at this normative mediated sort of normative like calculated way of getting everyone at like the gold directed Norm for a Westerner um and then can these different components to that be sort of treated and also looked at how do they vary as we move like not just as the five but IND you know as as individual kind of um tap ways to tap into affordances when do these change more or less when we bring people into another type of way of a regime of attention or another set of affordances which is outside the normal for our cultural go driven way of being in the west okay Colin and then Marco I just had a couple of thoughts about phenomenology since that's been a a theme um recently um first of all I think one interesting thing to think about in relation to uh trait approaches is that when we

do assessments of Personality using self-report questionnaires you know which is not the only way that you can do it of course you can have other people you know rate the personality of somebody and we tend to see very similar results if we use other people's ratings versus self-ratings um although there's also evidence that there's some incremental validity there in other words you know things about yourself that other people don't know so that there's some unique validity to your reports but other people also know things about you that you don't know so there's also some unique validity to other people's reports on your personality but as long as we're talking about the self-reports then that is at base a kind of you know crude phenomenological measure right because you're saying what are you like what is your experience of your s like are you the kind of person who's anxious are you the kind of person who likes art you know and we get a long list of people's responses on these things and then we uh infer their General patterns of uh behavior from them and so I do think you know one of the things that I've always been interested in is that personality is not just patterns of behavior from a third person perspective it's also patterns of experience from a firstperson perspective and so when we look at the contents of Personality questionnaires you know they involve descriptions of emotion and types of cognition and patterns of motivation that can all be experienced you know phenomenologically in the first person as well and so we're using this kind of first representation and understanding of ourselves to then try to infer these General patterns of behavior that have validity from an objective third person perspective as well and then just in terms of like drilling down to this kind of micr phenomenology idea we do know things about particular patterns of processing that are associated with different personality traits uh so for example we just published a paper showing that extraversion is associated with the tendency to be influenced by uh reward in an implicit learning context right so you're doing a task where you're making this like discrimination between two very similar options um and you're rewarded more for getting correct responses to one option relative to the other well what you see is that um people's behavior becomes biased toward that option but the degree it becomes biased is actually correlated with extroversion and that's part of how we you know have support for the theory that fundamentally what causes all of these different extroverted behaviors to cohere is sensitivity to reward um and so you can think of that from the third person as like here's a process of you know reward um mapping but you can also think well what is it like from for this individual this is the kind of person who's going to be more uh perceptive of possibilities for uh rewarding inter interactions for incentives in their environment so I think you can always sort of flip back and forth between taking the

phenomenological and the more U third person perspective on these things cool thanks Marco yeah uh thanks actually um kind of ties into that because I was kind of wondering if you could share some thoughts on the relation between value systems such as pratic value or affect and epistemic sence and curiosity with this uh model of Personality for example I did read that you said the fundamental components are um the goal the interpretation and the strategy right um and and I would say that that value I mean I would guess that value is more about the interpretation right so that you have some kind of scaler or some other dimension by which you can compare certain current real States and expected States um and tying into that uh I guess it's a separate question but I think I guess I'm going back to the personality and phenomenology because maybe you could share some thoughts and to what extent personality can be seen as an as a natural consequence of the Cy IC imperative where conscience has basically instrumentalized to be the regulator of uh entire mind body system as it were um because I guess I have some image picture of uh reses of reflexive attention right so there must be some cyb active system or set of systems ecosystem if you will um that's regulating or controlling the entire mind body system or large portions of it um but that in itself then has has to necessarily buy free energy or back of inference have to have some kind of um uh imperative to decide or policies to pick on how to engage with that bigger mindbody ecosystem if that makes sense beautiful thank you Adam and then anyone else darn it Marco uh that was awesome yeah was that was epic it was epic Adam and then Sasha and then we'll leave the warm-up so I guess uh two things Com or a few things coming to mind um so I really like this idea of thinking of um your Consciousness as a as a kind of modeling for the sake of cybernetic control and uh with hopefully you know some degree of not perfect but some degree of fidelity for modeling that which is modeled uh good regulator theorem to regulate a system you have to have a decent enough model um the and so there's a sense in which um towards this end uh being able to uh have your modeling process reflect on itself and model itself uh that that could be a very useful thing uh if you get it right um and a very bad thing if you get it wrong um the and so related to things like and it could even be that you know these parameter Ts for these they might stripe it across like multiple levels of abstraction something like curiosity so you could have like a kind of um uh curiosity that has a phenomenology in terms of like the way like a physicist would describe it as like a phenomenological description of behavior that's just like showing the emerg behavior and this looks like a more Curious system in the way it's like sussing out State space then you might also get to like uh curiosity in terms of the uh what it feels like to be curious and what those

those motivations those free energy gradients they going to try to annihilate those different sources of Desire of pull um what that's like experientially um and so yeah it seems like something like curiosity would be reflected at multiple levels of organization uh maybe even with some uh weird shifts in terms of like the way it shows up across uh levels of analysis um the GU a few other ideas are coming to mind would be um there seems to be a sense in which if so we're talking about you someone reporting on their experience um you can think of these reports both in terms of the uh perceptual inferences that are uh going into them and then the report is a kind of action as a kind of Act of inference and uh each of these aspects you know for in like you could set up a generative model for each of these and you would have for instance a given kind of uh you know Matrix to describe the perception like given a given set of observations what's the state that you then infer and so this could be like some aspects of like so what isn't agent what is a person uh noticing in different circumstances and so this is like what is what is it in particular that is entering into their experience where does their attention go you could also though have like their attention not just in terms of a sort of like uh a salience but a kind of like active curiosity and a a sampling from the environment as a kind of mental act and so this then will influence like what's getting fed into the stream of experience and what it's like to be uh a particular kind of agent particular kind of particular kind of person in a particular kind of situation um those are some thoughts cool we'll return to that idea about being in the niche for a particular kind of agent we'll do Sasha and then Colin with the last word on the intro awesome yeah thank you um really great conversation I've already reduced my uncertainty on a lot of the questions I came in with um uh than thank you Colin for explaining that um behavior is not the same thing as personality because that's something I was starting to wonder um so it's yeah relevant that personality uh contains as part of it the self-report of the um individual's experience um and then uh I like what you said about your most recent work that sounds really interesting and kind of Falls in line with um active inference uh as reducing uncertainty about our environment and so it seems that uh extroverts and introverts have a different sensitivity to um uh sens sensitivity to Precision about their world where extroverts are more sensitive to reward and um updating their generative model from the external inputs like mainly I'm thinking social but of course could be other things um so that's an interesting how it fits into into this model of of active inference as well and um I guess my last thought is is there another level up um here where we can talk about people who are also sensitive to Precision about personality tests and um others Quantified Self um whether there's a metric for that kind of like self-reflection part of it um

and if that I think some sometimes that can seem a bit like a self-fulfilling prophecy or like I know I'm the kind of person who does this so I expect this personality test to tell me this um where you're actually not sensitive to new information you just want uh kind of confirmation on what you already think so but yeah okay interesting stuff Colin and then we're going to move to the next section okay so quickly then just a couple of thoughts on the last couple of comments um so first of all you raised something Sasha that is important which is that personality tests are certainly fallible right they're not a great form of measurement they're just the best form of measurement we have for assessing people's personalities and the reason for that is that it's really hard to get at an assessment of people's persistent patterns in all of their you know different aspects of functioning uh with anything other than relying on people to tell you about it right you could try to come up with you know like some complex set of tasks and people come into the lab and do all these things and you videotape them and somehow that's going to you're going to extract all the information about their personality from that by a carefully chosen set of things that you have them go through you know it'll be a huge amount of work and you'd be learning less about what they're like in general than just asking them a bunch of questions so you know this is always a perennial problem it's like well there are these difficulties with self-reports people have biases they uh you know they are motivated to think of themselves as being more desirable in various ways than they are from a you know social desirability perspective uh and so what what can you do about that and yeah there are certainly studies of you know the degree to which people are accurate in their personality assessment and like the degree to which if they assess their own personality how much does that agree with other people's assessments of their personality and larger discrepancies can be predictive of things sometimes of psychological problems um so there's a you know there's a whole area of studying that um but I think I like this idea of people who have curiosity about their own personalities um and that kind of leads into something I just wanted to say generally about personality which is that definitely uh extroversion is related to this kind of curiosity about things that are potentially rewarding but if we think about curiosity more generally as a sensitivity to the reward value of information per se so just essentially the motivation to improve the quality of our interpretations of the world in terms of that uh language of interpretations and goals and strategies then we have a there's a pretty clear sense that Curiosity of that sort of general form is the core of one of the big five traits so the openness to experience and intellect dimension of the big five is really focused on people's curiosity uh and hunger for information essentially whether that be more about aesthetic and perceptual

information related to openness to experience or whether it's related to abstract information related to intellectual issues um so that that seems to be one of the really core dimensions of people's personalities so I do think there's a there's a very clear way to connect that into this idea of you know foraging for uh for information and seeking out increased entropy for the sake of reducing entropy uh over time you know which is what learning is essentially so there are good connections there definitely interesting and Adam just one quick last note uh so yeah quickly um guess before we temporarily close off the phenomenology thread um it's seems like there's a really uh Rich kind of biral relationship between uh personality science an active inference and that uh with personality science you actually have this like like they're all basically these um uh from the personality modelers I met they're all these very like integrative generalists approaching this hairball of a problem from as many ways as they can and basically bootstrapping their themselves to validity and reliability and so they've come up with uh a really rich literature of phenomena to populate the phenomenology and AC of inference provides the sort of like places where you want to kind of dig in deeper onto like well what is the exact like causal mechanism so like cybernetic Big Five theory was probably like best um impersonality science of getting at causal processes of like how you're uh what where are these traits coming from and what are they actually doing for the agent so now you can bring an act of inference and describe agents of cybernetic systems run simulations uh generate things that would correspond to different aspects of Personality like for instance the dopamine system and curiosity you now can modify your parameter over policy selection and see what then unfolds with respect to both over Acts or mental acts like attentional shifts um just one final thing before we go to the next part is uh there's some uh in the paper we start to talk about this idea of like a p struct um or like this kind of object that could be used in something like comput computational Psychiatry for um kind of pooling data over multiple multiple people to like come up with um basically a general personality model that you can deploy for different agents um we might want to go into the Weeds on that later um to the extent we can but the thing I want to get at is that uh the one thing to model an AC of inference is we are all personality modelers it's like we are all necessarily like super forecasters trying to predict the future we are all also personality modelers trying to predict each other and ourselves and part of this would be towards this end well what kind of agent do I think I am what do I think my personality is like what do I think is going to be either more situation like what's going to be enduring uh what's going to be you can have personality be enduring but variable so what's going to be uh how variable are these aspects of Me How

likely are they to endure these are the questions you have to ask of yourself and other agents in order to govern the interactions in the best of your ability well said Adam it really just highlights how much people even though we say things like well we don't know even given someone's personality we don't know what they're going to do it's like yeah but you know what language they're going to speak and so in the broad scheme of things you've actually reduced your uncertainty quite a lot and if you're not 99.9% aligned with that agent who's like you in the niche sharing an interface with you that conflict or adversarial relationship or just risk surface is going to spiral and so it's all about this interface yeah Marco if you want to go ahead yeah thanks U sorry I I didn't get get that far in the paper but that sound really interesting I just briefly skimmed the section on it so so it sounds like the P structure is basically allowing for a heris contextual dependency so they have the generic generative models of your being and the and the P structure kind of uh links the current context to the generic uh kind of models and that allows for the variation over different situations I mean I'm emphasizing theistic nature of it because personality basically can be seen as a very useful theistic for how to act in the world does that make sense or should I just read more in the paper no I think that's really a great uh connection there and a setup I think in figure two or when we see the graphical model maybe let's think about where this structure would play in and then emphasize its role as a statistical construct that reduces our uncertainty about regularities in the world just like every other kind of statistical modeling framework does um and then I think we'll be able to come back to that P stru question but what a fun uh section good times so in today we did the introductions and warm-up and then we'll have the sections of this paper which maybe if you're listening you've read or not and that is the integrating cybernetic Big Five Theory with free energy principle a new strategy for modeling personalities as complex systems by saffron and D young both of whom are on this call so in this set of slides we do have a bunch of uh the abstract road map some other stuff but I think there are so many interesting threads we've raised let's just try to walk through all of them um in no specific order but really think about what topics do we want to highlight for next week when we talk about the same paper and how will we get deeper into some of these threads that we've raised today in the discussion so here's the ases and claims slide and Adam or Colin um what was just your intention with the paper and after doing it where do you see the next implications of the paper or the next research directions for what you kind of want to do in that direction and if that verges into some of the figures just me mention it and I'll go to that figure um well I think the our basic motivation for doing it was just that we had already been talking together

about topics of mutual interest and the opportunity to write this chapter came up and I thought oh this would be a really fun thing to uh a good excuse to do something with Adam and kind of get our thoughts together and you know actually written down and out there in the world um and so you know I think he and I probably have slightly different incentives for where this is going next um he's more interested in the process processes of modeling um whereas I'm more interested in you know generating knowledge about the specific mechanisms and uh you know brain processes that underly different personality traits and risks for different types of psychopathology that are tied to different personality traits um but um yeah I think I mean to me this is just a really nice way to kind of connect U my thinking about personality with this uh what seems to me to be a really generative perspective with the whole Fe framework and what's happening with that in science in general I'm sure Adam has some other thoughts about that too yeah go for it Adam but that was very interesting Colin thanks oh I mean for me that was primarily the thing like col and I were already kind of going into the weeds together and having fun with it and so it's then uh basically uh uh iarpa brought together this meeting of personality modelers to create this book um modeling persons and situations um basically seeing the extent to which personality science could do uh predictive modeling of Agents um basically they want to recapit the plot of westw world as far as I can tell um so I'm actually not joking but except I am uh but yeah so my primarily uh uh interest would be just uh what the heck our persons uh and uh what can they be and how can we understand them and uh who am I and what do I want yeah and also one thing we haven't raised is who or what is a person there's personality as being the statistic iCal regularities of agent in Niche so that's just uh like a person in an exoskeleton you could have some sort of model for that or a person with other types of constraints on or something like that but then there's the idea of like what is a person legally what is a person culturally and all these ideas are coming into play with modeling legal persons and physical persons and regul statistical regularities that include our digital affordances all of that is becoming integrated our digital Persona is that included in these uh emotive models there's just a lot of really interesting stuff to think about there so then just copied on the screen for people who are looking is the abstract you can pause the video there and here is the road map then we'll have the figures and then there's just one kind of interesting general question from an audience but in the road map just if you have any thoughts on the layout or the authors could give a point if they wanted to about why they arranged it the way they arranged it or which parts they thought were important if somebody was just going to read one section or any thoughts on that otherwise we

can just go to the figures but okay let me know if you want to return to this otherwise it's just the headers of the paper all right so here's figure one and Colin this really returns to your point from earlier about the graphical models what do Connections in graphical models represent and that's very related to a general question of like what do variables represent principal component one what does that represent what English word are we going to put on that you know $MX + b$ linear model what are we going to call x what is the slope what is that going to be called and then how does that play out graphically whether in an equation or whether in a drawing like this so I would love to hear your thoughts and then anyone else can raise their hand on this one okay so I made the basic distinction earlier between a uh a St a statistical model of covariance like a structural equation model uh I think it's really important not to call structural equation models causal models because they're not that drives me crazy when people refer to causal modeling and all they're doing is a set of regression paths right what that is fundamentally is trying to U you know recreate patterns of covariance in data and if you have special kinds of data like with temporal priority or where some variable has actually been manipulated uh then you can actually interpret um you know causal effects from covariant structure but uh just in and of itself there's nothing inherently causal about structural equation modeling or regression um this model here however is different this is actually intended as a conceptual causal model so these are not just looking at correlations between different variables in a population this is thinking about the actual causal processes that produce the functions of a of an individual agent specifically a human being um and so what this is this is just sort of like a way to encapsulate an overview of the way that my theory uh defines personality traits and characteristic adaptations which are the two elements of exhaustive elements of Personality uh in the theory and so the idea is that you know we know from Decades of research that both genetics and the environment contribute to uh people's personality traits and the uh idea is that personality traits are patterns of behavior that's why they have their own separate box there there's sometimes a tendency in personality psychology to say well what the trait really is is some underlying thing that causes people's behavior um but I think that that leads to confusion because what we actually measure when we measure personality traits is a description of regularities in the person's behavior and experience uh so I think it's important to keep that clear but then the the middle box there the relatively stable parameters of cybernetic mechanisms is trying to say something about where those patterns of regularity come from and so the personality traits are not some underlying causal process that generates personality they are just the patterns now they themselves do have Downstream

causal effects however because let's say I'm the kind of person who talks all the time well that's going to have consequences on the world around me right and that will maybe lead me to develop different patterns of adaptation to my environment to have different outcomes on the people around me it's going to lead to different outcomes and different adaptations so that's the final box there um and then there just you know there's identification of feedback loops here so the idea that you're generally speaking the flow is that your basic personality traits caused by these basic settings of parameters that have a strong genetic component to them influence how we adapt and how we you know what we pick up on and what habits we develop in any particular situations uh habits is a good synonym for characteristic adaptations um but sometimes there can actually be feedback so let's say for example that I'm an introvert but uh for whatever reason I get into a job that has a an important component is selling things right I get into sales somehow so I'm going to have to develop at least in that context the ability to act like an extrovert if I'm going to be successful in my job and maybe I get good at that right so I've got a characteristic adaptation to my particular Niche which is that at least in this context I'm good at acting like an extrovert but this acknowledges that there's the possibility for that to feed back and actually make me more sensitive to incentive reward words in general and so I may actually find that I'm not just more extroverted in this particular job context after five years of doing this maybe I'm more extroverted in my life as a whole in all these other contexts um so that's the one of those feedback loops and then so you see that there's a the environment can directly affect the kind of ways in which we adapt to our environment obviously but the environment um but genetics any effect of genetics on our specific adaptations must feed through the effects of genetics on these General mechanisms these parameters that underly our functioning um and so you know if anybody has any other questions about that diagram I'm happy to respond that was a great explanation and just to um emphasize one point there and then we'll go to Stephen you said that the characteristic adaptations in life outcomes are exhaustive and so um that really is like we can partition them into the adaptations or the things that the agent carries with them the synapses that have changed or the scars they bear but basically what is not their surrounding and the life outcome is the niche so then there's this stigma G where the agent is doing Niche construction so this is from a different nonactive inference non- Niche construction framework but finding resonances and finding uh concordances here because that's kind of what we talk about like the learning of agents and then the way that influences their Niche and then you had this idea of acting as if like the example with the person who's um being forced by their Niche to

implement uh I'm a kind of person who says hello in this way or things like that and then that also relates to what Steph was saying earlier about drama and about performance so maybe Stephen do you have a thought on that yeah I mean the the the thing I like about what was mentioned here is how personality traits can be adapted to context like someone's A salesperson I thought was a useful example and then how that obvious starts to then change your whole broader way of seeing things and one area I think might be interesting and I've been there's the work of Harry Hest who's revisiting this the work of Roger Barker from the 60s we did a lot of work on Behavior settings so he looked at a whole load of children across the US and he looked at how their behavior was when they were in a school or in the cafe or different places he found the difference in Behavior between settings for the same child was bigger than the difference in Behavior between personalities of children I the same child was changing its Behavior radically based on the settings um and that that work kind of got lost but Harry heft has sort of been really pushing that again I I think that can tie into this is maybe there's personality traits I suppose you you know how much stability there is but could Behavior settings be kind of like that kind of more Dynamic version personality traits and I think active influence makes that more possible to even have that conversation rather than the more sort of static psychometric data which is kind of topped down and kind of you just have to accept it feels like this is M you can get in there and play with it a bit more from different directions thanks for assuring that Stephen we'll go Colin and then anyone else who raises their hand yeah so that's a really a really good point Stephen which is that uh there is there are differences that people have to specific situations that are distinct from their broad patterns of consistency and personality and that's a long running debate in personality psychology going back to the to the late 60s um and the resolution to it basically is that yes there are these relatively stable ways in which people respond to specific situations and that's what I'm trying to capture with the idea of characteristic adaptations but then if you go across a bunch of character a bunch of situations and you take the person's average there are also persistent averages that differ from person to person and that would correspond to the trait so you can have both of these things in the same set of data um andh will fleeen is a personality psychologist who has really been instrumental in collecting the data to show this very basic feature of what are we measuring when we measure a personality trait with something like a questionnaire where we're just asking somebody for this sort of general impression of themselves and so what he's done is used experience sampling methods where you like you have somebody's smartphone and it you know pings them like five times a day and asks them

like how have you been acting for the last you know half an hour or for the last hour and then you get them to rate their behavior on a set of uh you know on a set of descriptions or categories what you see is that uh as you said people have as much or even more variability within their own behavior as there is variability in terms of their average behavior from person to person and yet nonetheless those averages are quite stable so if you do this for one week you can get a whole you'll get a he calls them density distributions right you basically just get a you know and they're usually fairly normally shaped um and you do that for one week you get this whole pattern so sometimes the person's acting very introverted sometimes they're acting very extroverted you know at ranges um but there's there's a peak and that Peak is very stable if you do it again in another week so the person goes through another week of behavior and again they're varying very l in their behavior but if you look at their average level of that particular Behavior it's quite stable over time and so and then he's also shown that that correlates quite highly with the ratings that we get from these General personality questionnaires so the questionnaires really are tapping into this sort of General average tendency of specific behavior even though of course the behavior itself varies widely from situation to situation and over time both due to situational affordances and due to whatever is going on internally in terms of the goals that the person is pursuing at that moment awesome let's go Marco then Adam thanks it was actually really interesting I apologize I'm going to add another interesting thing to connect to so so so someone mentioned Steph mentioned the uh the kids in the classrooms and the context and I think development might be also very interesting here because um personality might be relatively stable across the lifetime but but I think hopefully we can agree that personality also enriches um and in in that process of enrichment and I would also say empowerment um there's I think you could say a shift between the balance of endogenous and exogenous influence on an agent's behavioral adaptation or to its to its niches um and so there's a there's a change in the scope in which the personality uh has a relatively large influence on the way they adapt and I guess I guess I'm just asking about thoughts on how you might picture that kind of process of how personality might actually in the beginning um largely concern itself with how to engage with the world such that it might internalize that which it normally co-ops from the environment so normally offload a lot of stuff to behaviorally adapt which might explain the the variation due to environments but then over time as it does so it internalizes certain heris certain patterns certain policies and learn from environments to kind of enrich its personality and then extend the scope uh of of influence from their personality um yeah I don't know if that makes sense

but yeah cool thought Adam and than anyone else uh so few thoughts from before and then um some thoughts on Marco's thoughts um so the free and description of uh personality as a kind of a density uh distribution um I think and the issue of um the stability of personality and the variance of it um with within individuals across context which is part of the reason the uh book that was this chapter was part of was called modeling persons and situations but uh taking these ideas together uh we're kind of in the same territory uh that of these fundamental questions in the free energy principle of like what is the ontological status of a non-equilibrium steady state distribution bounded by a Markov blanket um what's the granularity over which we can find this this thing that we're identifying so it's you're uh and so um col will be able to speak more to this but this idea of like there are ways of describing that uh your degree of variability across context is itself something that can be stable and this could be itself part of your personality um to uh now kind of loop back uh to what you were just describing Marco um I really like that idea um so it's like you're you're starting out in a situation where through this interaction between you and this context um that you're entering into in certain ways based on your own expectations and the expectations of people in the environment and and the structures that are there built up from this there's a certain stability and then you Bo you uh this stability um ends up shaping you um both just as as a consequence of like this is the thing for you to predict but also in the active inferential sense and as you're saying empowerment this is what's going to help me get better purchase I'm reducing my uncertainty with respect to realizing the value of uh existing and so this um idea your personality being kind of um internalization of structure and maybe at a certain stage almost like a kind of amortized inference um in terms of I got from you uh yeah I really like that idea a lot cool Adam Shannon you're muted ch thank you um going on this developmental thread and um the paper that you brought up this morning Daniel and um Anna's newspaper on the the first priors when you're in ruro as a baby I haven't read it yet but um the idea that like when you're in a neonate in a body you're literally relying on someone else's interceptive States and biological states to regulate everything about your inceptive in biological States and then you're born and you're still kind of relying on um that caregivers inceptive States or their behavior when they're holding you or coming to comfort you um this I think could relate to what Marco was saying about how your personality develops so first if you have if a personality is something like a collection of effective States and behaviors and an infant is completely sort of reliant on the environment and the social environment to regulate its own personalities and as you grow through toddlerhood you slowly are able to control this yourself

like you can maintain your own body budget a little bit more um learn how to how to get your own blanket if you're cold or like self soothe and then you grow older and maybe you're taught how to make music and learning that this is a new way that you can explore your personality or create it um and I don't study development at all so sorry if this is wondering maybe Sasha can pick up on it but I I feel like this can kind of connect if instead of looking or not instead of but in addition to looking at averages but looking at like a Time series of personality development over time and like you could do at any point in time or at any range of time some sort of um like what would be the the active interests model of your life in infancy in toddlerhood in you know Rowdy teenage years and as adult and you'll have you you'll find these like steady trajectories at the same time that a bit R Hope some can pick that up let me connect that to a few other points So Adam raised this question of what is the level of the thing or how do we identify things from empirical data specifically so that we can use it in active inference type or free energy principle type models do we use renormalization group or principal component analysis or some type of adjacency Matrix some type of marov blanket what are those things that we're actually going to be modeling and that reminds me a lot of a theory called ant uh actor Network Theory by Bruno Lor and then also on the ant wavelength Colin was mentioning how there's so much variation Within individuals but also St stable differences between and that reminded me a lot of my graduate school work with ant colonies where day-to-day the variation across all the ant colonies was extreme if it was raining they would do one thing if it was very dry versus if it had rained the day before those were massive determiners of colony behavior however across days and even across years there were stable markers of colony behavior and there were characteristic ways in which they responded to to the environment so it is possible to have a type of measurement and understanding of a system which my adviser established through long-term observation where you can actually model this development through time as well as within a population which doesn't erase individual differences it actually is the framework by which individual differences arise and then Shannon is so beautiful with music being introduce like a thread of personality and then that can be used to explore these regularities Adam then Stephen hi um what was the paper that people are mentioning uh about like first priors I'll post it in this jitsy and we'll put it in the YouTube chat but let's continue without worrying about this paper okay go on oh sorry should I go do you want to can I I just I was just going to put in one thing maybe I just um because I like when it mentions here relatively stable parameters of cyber cybernetic mechanisms sort of those kind of Dynamics and I think that links also to this idea of how that might change during

our life and thinking about our lifetime and one thing that I'm very interested in is when trauma comes into play PTSD and other types of situations and you know it's um it can really shift the state of someone and poly vagal Theory kind of talks to that where we've kind of got this sort of social co-regulation that we have as humans but we also have our kind of Primal kind of fight or flight or gut sort of sort of fear um responses and how those things are in a way they another type of like patent Behavior which becomes incredibly stable too stable like you can shut down or you might get into rage but then over the period of the day or life the challenge is they may it may be unregulated about when they appear so someone might sort of be a certain way and then they flash into rage you know and then they come come back into and their personality is radically different and I think this is so there they might be interesting to bring in that that that as a as a kind of a what happens when that relatively stable State itself flips you know um so the and it so your your your mechanism might not be cognitive but it might be in that kind of um polyal system yes the body influencing phase transitions in personality or behavior Adam and then Colin and then Marco we muted ad thanks um yes I think uh Colin should probably speak to what was just brought up in terms of like what actually happens with people um but in terms of uh like a high level of abstraction um so yeah it seems like there's this sense in which you can think of Personality as a kind of core screening but it's and a kind of renormalization and um in terms of it existing as a phenomenon and it's a and epistemically it's a kind of core straining um afforded by this renormalization um the this kind of throwing away a variability as you form these higher level attracting States um that you can uh ignore them uh with within the scope of a certain type of modeling that you're engaging in with respect to having certain objectives of trying to predict certain things um whether you're a personality modeler trying to predict someone else or trying to predict yourself or whatever um the So within this um looking for this optimal level of epistemic grip in terms of like the granularity of your description um this there you could describe these cybernetic control parameters as being uh part of the causal process of what's responsible and then if you take the system and you perturb it there's going to be like a range where like the order parameter changes and you can get these nonlinear maybe phase shifts where the um you're now in a different regime and it may look like a very different kind of person because the personality structure as this stable attractor Network um can then become destabilized um and so that would be like a high level in colge probably speak to like what actually happens with people cool yeah call and go for it so you get these phase shifts in Behavior but they're often due to persistent sensitivities in the underlying mechanisms right so

let's take um well so uh PTSD okay um what you have there essentially is some kind of traumatic event uh is an intervention that resets the sensitivity of the threat systems and so the that corresponds quite well to the big five dimension that's labeled as neuroticism which is an old-fashioned word but in this context basically just means sensitivity to all different kinds of negative emotion um and so obviously that's heightened in PTSD and what that means fundamentally is that you've gone through some kind of traumatic event that led to your your it wasn't just learning about a specific event right what it did was that it reset the sensitivity of your threat system as a whole so now you are perceiving threats and reacting with defensive negative emotion to much more minor situations than you would have previously um and of course even in terms of risk for PTSD we know that people who are higher in threat sensitivity to begin with are more likely to develop PTSD so it's not like it comes out of nowhere typically but nonetheless there obvious vious ly this change um and there was a a recent really nice meta analysis of Personality change following uh therapy and different types of psychotherapy whether it's pharmaceutical or talk therapy or cognitive behavioral therapy all of these lead to changes in personality and the the trait that changes the most as you might expect is reductions in neuroticism right so uh what we're talking about here is the ability to shift these underlying parameters that are not constantly manifest in the same way so when we say that somebody's high in trade anxiety for example we don't mean that they're experiencing anxiety at every minute of every day at a high level right we just mean that they're more likely to be anxious more intensely in more situations than somebody who is low in trade anxiety but nonetheless then if we really want to understand the Dynamics of the system we need to understand those phase shifts from like feelings of safety to feelings of anxiety at any given moment and I that's what you're talking about but in order to understand how that works for a given agent we also need to understand these persistent patterns of the sensitivities that lead them to have these shifts in different situations and to different degrees than other agents very cool thank you Marco then blue thanks um yeah this is really interesting so um trying to tie back to the earlier talk about um the role of a effect to this aspect of trauma so I think it's mentioned before by by the two authors that that um we could uh validly see these complex systems as composed of uh different micr personalities distributed over scale and levels um and and similarly for affect in the sense written about by KAS per has and others so um Adam safron Adam also mentioned renormalization right so so I think that ties in also to a lot of different things like Stig or equilibration and I think perhaps it's it's a had a thought or don't want to share is maybe maybe um the clustering phenomenon of personalities might be a

natural consequence of something like renormalization or equilibration if there already is a certain heterogeneity if there already is a certain kind of a symmetry in certain factors certain basic mechanisms of the CTIC uh systems shed throughout uh then that naturally should cluster because then they are more likely to cooperate if it remains heterogeneous and disordered or not co-organized uh then it's inefficient but then you have this thing with trauma which I think would disrupt that process right because if you go back to the component you pointed out interpretation goals and strategy then that's basically the key things about a trauma right you're in a certain situation that just demands an extreme change in your interpretational beliefs or models your goals for the particular situation and the extreme pressure to have the right strategy to get out of it and I I think maybe the the the side effect right the byproduct of that momentary extremeness is that it basically expands its influence and so it it becomes hypersensitive by nature and so uh the ongoing the always ongoing process of equilibration of everything co-regulating um becomes disrupted by that one participant the traumatic complex um if that makes sense so instead of a lot of different not these kind of equilibrating with each other kind of polycentric it becomes a monocentric equilibration if that makes sense it's AIT rambling lots of very interesting points thank you blue and then Adam so just to kind of take the trauma aspect maybe to the other side of the spectrum and and just in thinking about um you know like our own perception of our self right like so how we hold on to our own like person person ality versus is it you know congruent with what others think of us so I've thinking you know as how does this really relate to um meditation right like so in in meditation there's like the idea of getting rid of the ego right like so where like that's the thing that builds the ego or like the construct of ego so as we like eliminate this like construct of ego and also like staying in the moment right like we're not necessarily trying to build an accurate predictive model in in a meditation state so so when we have this like it's like this dissolution of of like free energy or or or what is happening like in in the meditative State um that that can also bring about a drastic personality shift I think as we get rid of the ego and learn to be non-reactive as opposed to like traumatic reaction that's nonlinear acceleration awesome Adam and then Marco then Colin and then anyone else who wants to raise your hand hi um so I I just try to address a couple of the threads um come to meditation uh at the end I think but the um it seems like a lot of the um principles of uh in terms of their generality uh could be really interestingly modeled um actually using things like ant colonies uh like in Daniel's work um and you might even find interestingly like even similar mechanisms at play so like in a recent paper starting to do active inferential modeling of ant Colony uh Behavior Uh decisions and

foraging um Daniel mentions like you know there's like the was it the foragers have more dopamine so you could see things like um the functional significance of dopamine actually as it's way of parameterizing a cybernetic system as being potentially similar although potentially switching across scales where in the individual ant and maybe even in in the personality of the colony as a whole and uh there's would be actually this interesting interplay between the personalities of individual ants and the personalities of colonies and this interplay which might not be an exactly a one for one might actually even be a kind of um uh complementarity where certain characteristics being present on um this broader level this this more encompassing marov blanket where the individuals are part of a superorganism or just the organism um from the perspective of natural selection because the colonies that which is selected uh as Daniel says um but so the person out of the colony let's say um would then interface and interact with the personality of the uh the individuals of which it's composed and uh potentially in an in training way but potentially not um to loop around um you might even be able to uh at the level of of colony talk about things like the aect of tendencies that it's you know it's what is its what is the affect the veillance the emotions experienced by the colonies is a whole in a given instance what is its um uh General mood and then what are its patterns of mood across different situations it's and you might now we're kind of going into what we more call personality territory and you could even think of things like um to kind of bring in some of like Marco's idea of the nature of trauma and this like dynamic system sense like you could traumatize The Colony and does it then at some point like there you would expect some sort of like elasticity or um what's the term like Hyer I don't know not that but it would be you perturbate and then it it would have some sort of adapt potential to come back from the peration because that's what we're trying to do is H keep existing in the way we are and so some adaptive potential but you push it too far uh and then you might not get back this um adaptive attracting structure you might get stuck in a different regime um and you could think of the same things uh similar things this could be a way of describing things that happen within the within uh individuals like so there's um you can think of that maybe on multiple levels of abstraction but stop for now thanks yep I just switched the slide to figure two so Colin and then Marco go for it and maybe we can introduce the p struct idea or just how does active inference specifically help us address some of these questions but go ahead call in and then Marco Okay well what I'm my thoughts are about a couple of the last comments so they're s more related to the to the previous slide and then go ahead then it's fine um so in terms of both this discussion of what's going on in trauma and what's going on in meditation I

think one of the interesting things to think about again is the distinction between the personality traits and the characteristic adaptations um because in uh in both there's the capacity for both of them to be shifted so uh Marco was talking about the way in which in a traumatic event uh you basically pre-existing goals and interpretations and strategies are dramatically invalidated and and that uh you know then means that the system doesn't know how to pursue its goals in fact it may not even know what its goals should be and obviously that's extremely disruptive and disregulated and so one of the challenges in the immediate of course is to figure out uh what you should be doing what strategies you should be using but then there's also this capacity for that traumatic event uh to shift the underlying sensitivities of the system to what is perceived as disruptive in future right to sensitize to to things that disrupt um and that is you know obviously an adapted response to say like if you're in a in a dangerous environment you should become more sensitive to uh to danger and the traumatic event just basically taught you that the world was a much more dangerous place than you thought that's part of what is uh tragic about it right because you have this persistent sense that the world potentially is much is much more dangerous even than it is or that you should experience it as being but then you have the same kinds of Dynamics going on in um in meditation where trying to reduce your reactivity that's like trying to shift some of these basic parameters of the system so that you you know react less easily you're disrupted less easily and then the kind of non eye idea the recognition of the the elusory nature of the eye what that's about is basically not being overly attached to your uh model of yourself right because the the model is a model it's not yourself so if you are overly attached to your model then you have you're overly attached to your conscious understanding of what your goals and your strategies are and what you're like and meditation is basically trying to get you to recognize that those ideas that you have a they're just a they're just your model they're not actually the way that you are and B they're contingent on your situation and so you should be willing to let them go as your situation changes and as things evolve and so it's essentially creating both a pattern of Greater flexibility in the capacity to adapt uh but then also the attempt to uh downregulate the negative reaction to disruption right because if you have more confidence in your ability to adapt then you should uh be less defensively reactive to disruptions because you can be confident that you'll simply adapt to whatever the the new situation is thanks Marco yeah thanks um yeah just just piggybacking on that because that was really nice um so so so my thoughts on meditation and very good and important question I think um so so going back to the image of having all these different distributed micr personalities or sub

personalities I think meditation can be seen as allowing them to manifest and arise as it often is called and manifest their the Expressions which are always going on to also reach as a were the global workspace or Consciousness or the uh conscious attention and so what you're doing then effectively is broadening the collective of sub agents or subpersonalities or subsystems such that that equilibrative process improves because then you have a bigger scope a bigger domain of possible inputs to an ongoing process of inference um and importantly this is then of course without need for explicit modeling you're don't need to explicitly label a certain um rationally or verbally explicated belief right it's an exercise in in in the in the deepest sense of it uh importantly it's it's it's effectively a reflexive Niche construction because meditation is systematically to basically dis attenuate to yeah to attenuate external inputs you sit in a very neutral position you you're in a calm environment external input is minimized and so all the inputs well it augments the inputs that is endogenous coming from within which of course um is then going to translate or transfer to normal waking States because those are all the endogenous or it concerns all the possible indulgent influences on your ongoing cybernetics and so it's it's it's a reflective shift right and yeah I don't know what I was going to say uh yeah so it's basically just a seric game so from what I have seen my limit experience with meditation and reading upon it it's it's all systematic cybernetic games it puts you in a situation that that forces you to train the process that underly sonetics concerning the purely reflexive which is of course beautiful because there's almost no opportunity for that it's an extremely exclusive state of being that exclusively uh forces you to attend to these indous processes it's kind of the sbics of the sbics if that makes sense great points there it's why so much meditation begins with thankfulness and just to give actually one behavioral example um let's think about the example of a circus elephant where when it's young it has a heavy chain and it can't break away and then when it's older a simple thin rope can keep it connected so at a first pass one might say um well why doesn't it just break away why doesn't it engage in a policy of breaking away and the habits the E the priors over the policy the field of affordances may not even include that as an option so if certain types of actions are down regulated so to speak by 100x before any evidence comes in then it's going to be interpreted within a prior that downregulates or basically downweights the relevance or the likelihood of certain types of actions happening but also there's this element of the preferences where it may through whatever mechanism it's not an exact metaphor here the system kind of like a Stockholm syndrome situation the system can actually uh uh internalize a preference for certain States that may or may not be consciously available during regular States Of Consciousness that lead

into the implicit processing of everything within a framework that results in Behavior that's not aligned with something else which could be external Niche pressure or peer pressure or it could be discordant with one's own narrative themes and then meditation is interesting to think about as far as isolating these different types of parameters in the model and different kinds of meditation lots of different ways to do it Adam hi um hard to follow up on all that the um some the way the threads came together for me uh was it seems like across varieties of meditation and why would you do such a thing um and ultimately like to bring back maros point about empowerment um it's you want to be more capable in some way having a better grip on existence and a particular kind of way uh you want to be you have more you want to have um more degrees of freedom uh with respect to the way you're engaging with the world and there's a sense in which you can think of different types of meditation as either as allowing you to uh navigate through cybernetically uh free free energy Landscapes which can uh and and also adjust the nature of these Landscapes such that they are navigable so uh you can think of something like and to bring back in like something like trauma or something like uh the elephant getting stuck you can think of this as like a getting stuck at a local Optimum um of of a potentially Jagged free energy landscape and so if you can uh let's say reduce the uh some emotional uh regulation capacity the ability to defuse from an experience and keep the gain on positive feedback loop cycles of thought and emotion from getting as high as they might uh you might be able to make the free energy landscape less Jagged the difference is from point to point it's going to be uh you're not going to have these wild swings and so this might give you greater latitude with respect to pursuing more complex uh policies and actually I hope at some point we start to discuss some of colins's work on well-being because I think it it docks uh there's a lot to be said there um but uh one thing uh on this one podcast con was mentioning like you can think of one of the benefits of mindfulness uh which is like in pursuing any kind of complex goal um there's going to be a sort of trading off you're going to have to do against different things on on road there and so if you can like temporarily uh let some of your characteristic adaptations slide um let some of your uh policies like not be as important if they're if they're not being you think they ought to be realized but they're not within some sort of like hierarchical policy selection well then you can pursue uh you can hit targets you wouldn't otherwise you can you're a mountain car you can get to the top of the mountain um and the mountain car problem um so yeah nice cool Colin and then anyone else I just wanted to say to point out that that's that's the essence of non-attachment right you're not attached to particular policies and that means that you don't get hung up on a subgoal

when you should be shifting policies because what you are trying to pursue is some larger goal there's sort of you know there's The Stereotype of uh meditation at least you know Zen meditation for example Buddhist meditation that you're eliminating your goals right you have no desires but that's not that is a that's a caricature right because it's not that you have no goals it's just that you're not attached to particular low-level goals you might be uh you know you probably remain attached to if you're a good Buddhist for example to eliminating suffering for all beings and so it's not like you're not go directed and you know you probably maintain maintain life by eating and things like this it's just that you're not attached to the particular policies that might be using uh to achieve these higher level goals because goals are are just nested I mean there's a way in which there's a goal but then the policy is also just a goal it's just there're nested goals all the way down to the terminals which are specific motor operations or specific cognitive operations and so non-attachment is basically becoming more flexible in which sub goals you are willing to consider and to use very nice blue so I wonder if it's like it's more that it's not attachment to the policy but that actually is the policy and it's not attachment to the specific generative or predictive model I wonder if that's like because the policy is then the policy of non-attachment to the model or or Ultimate model flexibility right you could call it a metap policy metap policy right and and so um and like secondly I wonder about uh you know the idea of staying in the in the present moment is this like some kind of free energy equilibrium like in and not necessarily minimization but like if we take the generative model and the predictive model if we take like the value to zero are we like is meditation like staying in the present moment just like this ultimate state of of equilibrium in in free energy like infodynamics well interesting point about the hyper prior the basian hyper prior and also that's why a lot of meditation processes involve rhythmic or the cessation of other inputs because it allows the interception potentially to proceed in a different way isolation tanks or uh other Altered States Of Consciousness so let's do keeping comment short because we only a few minutes left Adam and then Marco in terms of the there's this one paper actually that just came out recently about a model of meditation with respect to um the temporal depth and counterfactual richness of uh your generative modeling that just came out I'll try to find that and send that but I think that would speak to this idea of staying in the present moment also this idea of um staying in the present moment um also makes me think of this uh Mark was like kind of intimating that you can think of meditation as a kind of uh like game task or almost like a kind of like uh maybe maybe play even uh but it's this building capacity um that is building capacity okay we'll come back Marco for okay uh so so a building

of capacity through relaxing some of your priors and so letting um you to be uh informed by uh policies with respect to mental acts that you might not normally deploy and you might not normally notice um what is entailed when you deploy certain types of mental act so things might come to mind uh that wouldn't normally uh that would place you now in into a different state of uh affective inference and so I think uh if we wanted to do uh modeling of this of something like meditation uh we would want to use this sort of imaginative policy selection described by something like sophisticated um effect of inference and that would begin to get us there um the exact way and one more thought before we move on is um one more thing I I was actually wondering with respect to meditation I've talked about this some with Colin is uh above the traits in the personality hierarchy you have these meta traits of stability and plasticity and um stability uh is this shared variance over and uh Colin actually named these um it's the shared variance over neuroticism inverse neuroticism agreeableness and conscientiousness and plasticity is the shared variance over extroversion and openness intellect and interestingly the stability seems to be basically the inverse of the general Psychopathology Factor P um by and large um but I I think thinking of mindfulness and meditation with resp back to um building capacities for both stability and plasticity which um helping to U um basically place this so you can think of stability as your ability to follow through on and protect your policies or characteristic adaptations and then plasticity would be your tendency to form new characteristic adaptations so like whether um so let's say you do encounter trauma um plasticity might be what makes the difference so stability would be how much it might make sense to the degree to which you're actually traumatized like you're very stable something might you know bounce right off you you're you're not traumatized but let's say you don't have a lot of plasticity and that instability is compromised plasticity is the thing that would let you update and uh create new strategies uh for being in the world and so they're kind of the state of dynamic tension in that they can like pull at each other um but they're both ultimately synergistic in that like you're not going to be particularly plastic unless you have this Foundation of stability and you're not going to be able to maintain your stability long unless you're metastable and so um part of with meditation mindfulness I'm wondering if one way of thinking of it is it's part of what's actually really letting you kind of like ride this Optimum of balance stability and plasticity and that could be one way of thinking about part of what um people get from it cool Colin yeah that's amazing that's it's a perfect setup yeah um I I I also fully agree with this theme of complementarity right so stability and plasticity sounds really it resonates with picity sorry openness and focus which is a big complimentary Uhl yeah

big polarity in meditation practices right so if open meditation just feel everything let it all come in ETC Focus meditation you know focus on one point one Mantra um and that's has a bigger connection to the the more well-known complementarity or polarity of exploration exploitation right but it's not about the polarities per se it's it's always about the balancing them Ma and for example Sussex had also a nice paper on that it's not about perfect thetical representations you can have some errors it's about how you balance everything off and so again we go back to the idea notion that that meditation can be games where you effectively practice and balancing that on cuz that balance is not something you can explicitly verbalize it's intrinsically preverbal because you can verbalize maybe the particular strategies or beliefs for Ability or for picity but to balance them off is more effective it's more flexible or fluid and so the the another interesting maybe Duality that that you can think of is also equilibriums and games or contexts right so in Game Theory you have an equilibrium for a particular game or setting and so what you said earlier about equilibrium and staying in the moment uh regarding meditation I think it's a very nice idea what what Adam already said is that um by being in that particular game which is that meditative practice um you will naturally by free energy minimization go to a particular kind of equilibrium but that NE necessitates change some updates and those updates might actually mean that your equilibrium seeking Behavior or your attractors will be different in the other context so the underlying assumption is that these particular changes in equilibriums or priors or attractors um are transferable so um yeah does that make sense as for your question on meditation presence in the presence yeah cool thoughts Colin I was just going to say a little bit more about stability and plasticity um I think that those can be seen as the two fundamental uh needs of any cybernetic or active inference system that exists in a complex or changing or unpredictable environment right so stability is in some sense just the the most fundamental requirement because the system must be able to pursue its goals effectively in order to be a cybernetic system right if it's an organism that's what it must have in order to persist and even an artificial system in order to persist as the system that it is it must be moving effectively toward its goals um but then if you have any degree of unpredictability in the environment you can't just be satisfied with the same set of policies that you have at any given time you're going to have to be able to update your policies generate new policies and so then immediately in order to remain stable over time in a changing environment you also have to be somewhat adaptive or plastic and and so you have these uh inherent Necessities for the capacity to maintain a stable order of goal directed functioning and then the capacity to generate uh new components of the model right new

policies or even new goals or new interpretations of the of the situation that you're in and then those so those are complimentary but they can also they are also in Dynamic tension as Adam mentioned because if you're massively exploratory it can be difficult to remain stable because you have so much new uh you know information coming in um similarly if you are well I mean we've essentially already covered what the the danger of being too inflexible and too unexplored as things change over time then you become unadapted to your situation so stability and plasticity aren't opposite to each other right stabil the opposite of stability is instability the opposite of plasticity is rigidity you can be uh stable and plastic or stable and rigid you can be unstable and plastic and unstable and rigid cool Adam go ahead um it seems like at some point it could be good to discuss connections of um narrative perspectives on personality because this idea of um narrative as active inference and the work trying to be done there I think there could be like a rich interchange because in some ways you can think of know selves as centers of narrative gravity and you can think of person as um styles and modes of narration and themes and motifs as you are this semi-reliable narr narrator telling your story and making and enacting it and making it so to degrees um and this being a source of to the degree you have stability and plasticity um this would be one source of it the ability of narratives to provide cross situational intertemporal common context and but also because of their richness to create this um Rich this kind of their combinatorics and Rich representative capacity also being a source of plasticity and so um narrative being a source of balance Dynamic balancing this Dynamic tension although the narrative itself can also not be a source of balance but imbalance if the style of narration becomes problematic so that kind of mindfulness enters in there again where um it might like your usual styles of narrative narrative I ation you might let that go a bit you go um this is this is the way in which you go naive Bays on your expectations um yep the stories provide the polls and the archetypes and then you can say okay it's like Goldilocks but for the internet of things or something like that and that gives you a vocabulary to communicate and design you might need a very different story in a very different context and kind of bring it back to trauma that might be all the difference between whether you get stuck or you experience post-traumatic growth yeah cool interesting and just um one closing uh slide that I want to put up while people prepare their thoughts this is a question just from uh audience member they wrote I have a question about what happens when blankets intersect blankets of blankets and so let's think about this in this kind of blankets of physiologically relevant States informationally relevant what what um think about what we're talking about here and at which nodes that would happen and then

there's a link to a talk by Carl Friston and a question about how nodes intersect and just while people are thinking about this or their last thoughts on the paper uh one note is about the scientist and the personality scientist as an informational forager as an active inference agent themselves and so our quest to reduce uncertainty about personality it shouldn't be about the optimal descriptor it should lead to a question of ourselves which is what is the optimal perturbation that we can design that will reduce our uncertainty the most about this individual's and so that takes into account population level Dynamics as a prior but personalized and um kind of dynamic testing so imagine having a baseline prior across knowledge architectures that people had and then having a dynamically updating online course that was optimally informative from both sides of the interface it was optimally informative for what kind of learner the person was and they were reducing their uncertainty optimally about the topic at hand and so that was like a co-evolutionary process amidst a prior that was population level but personalized in the individual trajectory so it's a really interesting framework in col and Adam thanks a lot for coming on the stream was like really helpful and we're excited that you'll come back for next week so in these last couple minutes anyone can raise your hand give a last thought what would be something that we would want to say to this slide about how nodes intersecting stories intersecting individuals intersecting how does that come into play and or what should we be thinking about for 13.2 that would be good to cover Shannon and then anyone else who raises their hand I guess I have a thought about um the question on this slide um so is um one node that's a sent state in one blanket the internal state of another blanket like that's totally the case because you basically choose a perspective and whichever you know blanket sort of internal State you want to describe or whichever thing that you're calling the agent separate from its environment then that's the internal state that you're going to describe and if you're looking at like if this blue node is a neuron and you want to actually know about this pink node this other neuron's internal State then that's going to be your internal State and these other guys are going to be the sent State the blanket around it and that's going to be the statistical boundary that you're using to explain the behavior of that neuron and then you can Aggregate and this collection of neurons will all be an internal State for the next collection of neurons so the answer is in the question that's exactly right perfect answer showing that the answer is in the question is the answer Adam uh in terms of uh loop around to a previous thought in terms of nested uh blankets um and where these different blankets might have different characteristics in their interaction um for personality uh there's some interesting work um colum might disagree on tight and loose cultures where you can

think of like the personality of the culture you're within and this tightness and looseness seems to correspond to this stability plasticity Dynamic tension this fundamental cybernetic parameters showing up on the group level and you in theory can get this kind of complementarity there where a tight culture might create more tight um individuals with respect to the degree to which you they let their their Pol whether they have a metap policy of letting their policy slide more um in different context but uh or instead of it being this sort of entrainment relation it could be a complimentary relation where the stability on the level of the group this broader um uhou attractor affords the fine grain the subsystems to actually have more latitude and vice versa um and I I imagine you might probably see similar things play out like with ant colonies um interesting point about multi-level systems and how they're associated with relative constraints in the last two three minutes let's think of a last you know if anyone who wants to raise their hand give a word or a sentence or few what is something that you took away or was exciting today and what was something that we could continue talking about in 13.2 and pick up where we left off Marco and then just anyone else who hand yeah I just wanted to thank everyone this was like really amazing this my first this this kind of stream seminar uh in ages so thank you is innovating and I think in general what to be taken away I think is is uh excitement because it's kind ofing how many different domains and topics and questions have been successfully synthesized in this big discussion and I think that's a testament to the power of these kinds of Frameworks of invasion or Active influence frean Energy C etics when it comes to understanding us people us persons personalities um as individuals as as collectives so thank you again for exciting me thank you well said and the group performance is uh only realized it's only in acted even digitally when the people show up and it's only relevant when there's the regime of attention and so there's so much you know fun and it's great to have you um multiple of you on your first time here that's always appreciated so any other thoughts in the last minute otherwise in the calendar invite is a feedback form so if you're just wanting to think a little bit more if you're live participant you want to think a little more about what we could cover next week just put it in that feedback form and then anyone who's watching it just leave a comment for something that we could address even if it's after 13.2 maybe we can swoop back and return to some previous comments but this was a super fun discussion and a big thanks to all the participants so really good times everyone and thanks a lot so see you next week